

Dang Ngoc Khanh Tam

Data Engineer | Data Analytics Engineer

 0947780167

 tamdangngoc0707@gmail.com

 github.com/DangNgocKhanhTam

 HCMC, VietNam

TECH STACKS

PROGRAMMING & DATABASE: • Python (Advanced) | Pandas | Numpy | Cursor | Psycopgs • SQL (Advanced) | Bigquery | AWS RedShift | Snowflake | Postgres SQL | MongoDB | Cansandra

DATA MODEL & ETL: • Apache Spark • DBT | Star Schema | SnowFlake Schema | SCD • Cube.js • Sematic Model | Microsoft Fabric

DATA ORCHESTRATION & STREAMING: • Apache Airflow | Apache NiFi • Apache Kafka | CDC | Debezium • Apache Spark

VISUALISATION TOOLS: • Microsoft Power BI (Advanced) • Looker Studio (Advanced) • Apache Superset • Metabase

CLOUD SERVICES: • AWS | AWS S3 | AWS Redshift | AWS EC2 • GCP | BigQuery | Cloud Run | Compute Engine VM • Azure | Microsoft Fabric

WEB SCRAPING & MACHINE LEARNING: • Selenium • Beautiful Soup | Scikit-learn • K-Nearest Neighbors (KNN) • Logistic Regression • Linear Regression

LANGUAGE PROFICIENCY: • English IELTS 7.0 (Speaking 6.5, Writing 6.5)

WORKING EXPERIENCE

Data Analytics Engineer

BE GROUP

03/2024 – Present

Ride-hailing and technology platform, offering multi-service platform including ride-hailing (BeBike, BeCar, BeFood), delivery, and digital services

Data Ingestion & ELT:

- Collected marketing data **via APIs** from multiple channels (*TikTok, Facebook, Chợ Tốt*) and ingested into **Google Cloud Storage (GCS)**, orchestrated with **Apache Airflow** to batch-load into **BigQuery** to enabled **centralized and automated tracking of marketing activities across platforms**.
- Implemented **Change Data Capture (CDC)** for OLTP database using **Debezium** to stream changes from **MySQL** into **Kafka**, then synchronized to **BigQuery** via Kafka connector to ensured **near real-time data availability for analytics and reporting**.

Data Modeling & Warehousing:

- Designed and built **data models in dbt** for multiple functional departments (*Marketplace, Marketing, Operations, Customer Service*) to **standardized data definitions** across teams.
- Applied **dimensional modeling (Star Schema)** to optimize query performance and simplify reporting, while handling **missing data** and managing Slowly Changing Dimension (**SCD**) attributes to ensure **data accuracy** and **historical consistency**.
- Implemented **data quality checks with dbt tests**, scheduled validation jobs in **Airflow**, and integrated **job alerts to Microsoft Teams channels** that improved **data reliability**, proactively detected anomalies, and reduced downtime.

Monitoring & Optimization:

- Built **Usage dashboards** and tracked **data lineage** by integrating **APIs** from **Airflow, Microsoft Power BI, Google Looker Studio**, and **BigQuery logs** to monitor dashboard utilization and job performance that identified underused dashboards and high-cost queries, deactivated redundant jobs, and achieved **~40% monthly cost savings**.
- Designed a **Power BI App** to **centralize multiple dashboards** and reports into a single, organized portal, enhancing user experience and implementing **row-level security (RLS)** to ensure controlled access across business teams

Dashboard Development:

- Built and delivered **dashboards** for vehicle-related services (**BeBike, Car, Taxi**) in the **Be app** for monitoring of service performance and supported data-driven decision making.
- Collected requirements from business stakeholders and aligned with product owners to **validate business logic** and **design corresponding data flows**.

Skills: Biggquery, Python, GCS, Rest API, Kafka, Debezium, DBT, Microsoft Fabric, Apache Airflow, PowerBI, Looker Studio, Superset

Data Analyst

Onpoint Ecommerce

05/2023 – 02/2024 | HCMC

E-commerce Enabler, a leading company in Vietnam that provides e-commerce development support services

- Create **Customer Service Performance, Chatbot Effectiveness Dashboards, Fulfillment Tracking Order Dashboard... in Power BI** suggest key metrics to track performance. Helping the business team **save 25% time** in summarizing status brands and decision-making.
- Measure an effectiveness of Customer Service activities on a platform site (*Shopee, Lazada, TiktokShop, Tiki, Landing Page...*) and assisting the CS team in determining which key metrics and factor will **generate 65% traffic and revenue**.
- Optimize the **ETL (Extract, Transform, Load)** pipeline using **procedures in Postgres SQL and Airflow** has resulted in a significant **reduction of load time by 50%**
- Worked alongside various departments (including Customer Service, Operation & Warehouse, etc.) to gather data from multiple sources (such as SharePoint, Opollo, ...) utilizing **Postgres SQL and Python**
- Use a **machine learning model** to identify key factors that influence the effectiveness of customer service operations among the **16 defined factors** (Response Rate, Response Time, Shipping fee, Voucher...) with data sample of CS agents activities data continuously for 6 months.
- Collaborating with Product Owners to determine the optimal approach for **data flow extraction**, aligning with user objectives and technical requirements. This involves assessing data sources, exploring methodologies, and streamlining extraction while maintaining data integrity and efficiency.

- Implementing an **automated data export process** based on user-requested schedules utilizing **Python and Airflow**.

Skills: PostgreSQL, Python, Apache Airflow, PowerBI, APIs, Linear Regression, Logistic Regression

Data Analyst

Bluecore Company Limited

09/2022 – 03/2023 | HCMC

Data Solutions Company have digital transformation solutions, apply technology to data analysis, management, sales and customer care

- Create **BI dashboards (PostgreSQL, Metabase)** to gain new insights and monitor key organizational and for the decision-making process.
- Design and develop analytical capabilities, reports, and **dashboards** needed to meet requirements.
- Ability to understand business logic from end-user reports and procedures. Structure them as input to establish **data transformation**.
- Locate, identify and clean essential related data needed for **sales performance tracking dashboard**.
- Working closely with business users to define analytical needs and requirements.
- Collaborate with data engineers to develop and implement data analyses that optimize statistical efficiency and quality.
- Support customers by investigating and understanding the situation of the customers, **defining the solutions** and expediting actively.

Skills: PostgreSQL, MS SQL, Metabase, Google BigQuery, Power BI

Projects

Real-Time Data Pipeline with Kafka, Spark, and Airflow

Designing a real-time data pipeline with Kafka and Spark Streaming. Stock price data is streamed from a custom Market API and stored in MongoDB. A daily workflow, managed with Apache Airflow, extracts details about stock prices and loads them into a PostgreSQL database.

CDC - Sync Data from MySQL to BigQuery with Kafka Connect and Debezium

Building a real-time data pipeline by streaming change data capture (CDC) events from MySQL with Debezium and Kafka, then loading them into BigQuery through Kafka Connect to enable scalable data synchronization and support analytics use cases

Data Modeling with DBT (Data Build Tool) & BigQuery

Utilizing the **Wide World Importers dataset** on **BigQuery**, **DBT** performs essential data transformations (cleaning, structuring, and aggregating to generate analysis-ready datasets, etc.). The refined data is then stored back in **BigQuery**.

COURSE & CERTIFICATION

Data Engineer ZoomCamp – *Datataalk Club*.

ETL and Data Pipelines – *Dataquest*

Mastering DBT – *Udemy*

Advanced SQL – *HackerRank*

Data Visualization in Power BI, DAX Function – *Datacamp*

Applied Statistics in Business and Economics – *Dataquest*.

Advanced Microsoft Excel (VLOOKUP, Power Query) – *IDP*

Education

Bachelor in Business Administration

2018 – 2022 | GPA: 3.36/4.0

Ton Duc Thang University

- Mathematics and Statistics (3.49)
- Big Data (3.28)